

Predicting Next Session Lifting Performance: A Personalized Sequence Modeling Approach

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Abstract—In the context of strength training a lifter continuously aims to progress their strength. They must take into consideration the rate of progression with fatigue accumulation in mind. Getting this balance wrong can lead to a plateau in progression or injury. Existing heuristics in place such as standard linear progression [+2.5kg or +1rep per week] and the Epley 1RM formula $[\text{Weight} * (1 + \frac{\text{reps}}{30})]$ lack personalization. In this project I propose a sequence-based model (LSTM) to use a single lifter's workout log to predict their performance in their next session for the bench press and lat pulldown. This will be measured against the heuristics described above as well as a linear regression and gradient boosted trees model using RMSE and MAE. Expected results are a working performance predictor for strength training as well as an insight into features that have a high impact on strength training performance.

Keywords—Sequence modeling, LSTM, strength training, time series prediction, personalization.

I. INTRODUCTION

The current norm for strength training heuristics are standard linear progression [+2.5kg or +1rep per week] and the Epley 1RM formula $[\text{Weight} * (1 + \frac{\text{reps}}{30})]$. These are used as a generalization for every lifter and serve more as a general guide rather than absolute. These heuristics lack personalization and the ability to track accumulated fatigue for a single lifter. When misjudged, progression is greatly hindered and there is an increased risk of injury.

This project investigates whether a sequence model trained on a single lifter's chronological workout log can accurately predict the lifter's next session performance for the bench press and lat pulldown better than these heuristics, and whether it can be trained effectively on a single lifter's dataset.

II. RELATED WORK

A. Applying Data Mining and Machine Learning Techniques to Predict Powerlifting Results [1]

This study aims to predict raw and non-raw powerlifting performance. They do this by using a variety of data mining and preprocessing techniques with 500k lifters across 20k powerlifting events. The processed data is then used to train machine learning models.

B. Hybrid Transformer-LSTM Model for Athlete Performance Prediction in Sports Training Management [2]

This study uses 200 athletes over 12 months, training a transformer and LSTM with features such as heartrate, speed, distance, workload and recovery indicators to create an athlete performance forecast. The main model is then benchmarked against a just transformer model and a just LSTM model.

C. The Gap

Both studies focus on a broad population of athletes rather than focusing on just one lifter. The features used are also

based on either competition or physical signs rather than a day-by-day log where trends can be found for an individual's performance.

III. PROPOSED METHOD

A. Data and Resources

I was able to export my own lifting data from Hevy, a workout tracking software. This, however, was a limited dataset with only 128 separate workout sessions. To compensate for this issue, I have found a dataset on Kaggle called *Hevy App Workout Dataset: From Dumbbells to Data* [3] which consists of 100 weeks of training from a single lifter exported from Hevy. As both datasets were exported from the Hevy application, they both have the exact same set of features.

The relevant features are: title, start_time, end_time, exercise_title, set_index, set_type, weight_kg, reps. I dropped some seemingly useful features such as exercise_notes and rpe. The reason behind these exclusions are that I want the model to learn from patterns in the training not from what the notes about the exercise are telling it. RPE is also excluded as it is inconsistently logged in my own data meaning any patterns learned from this feature during pretraining would not transfer to fine-tuning.

My plan is to pretrain a LSTM using the larger dataset from Kaggle and then fine-tuning the model on my own dataset so that it can work as a personal workout predictor.

The project will use Python, PyTorch, Pandas, Scikit-Learn, XGBoost, matplotlib and NumPy. This will all be run on a personal laptop.

B. Feature Engineering

Raw log entries are recorded per set, meaning they must first be combined into per-session values. The top set (heaviest working set) weight and reps are used to represent each session for a given exercise, and this also forms the prediction target. Since progression and fatigue are muscle specific, features are computed separately for the bench press and lat pulldown. The weight used in the Kaggle dataset is measured in pounds while my own dataset uses kg meaning that units will need to be converted to kg. For each exercise, the following features are engineered:

- **Session volume:** total volume for the session, with sessions at or below two thirds of the average volume classified as low-volume sessions.
- **Days since last performed:** tracks immediate recovery time for that exercise.
- **Days since last low volume session:** tracks longer-term accumulated fatigue.
- **Progression trend:** the slope of top set weight over the last five sessions for that exercise.

- **Time of day:** included to explore, via feature importance, whether session timing has a measurable effect on performance.

All features will be normalized before being passed to the LSTM, given the difference in scale between raw weight, volume, and count-based features.

C. Model Architecture

The LSTM model will take the sequence of engineered per session feature vectors, which are used to predict the next session's top set weight and reps for the bench press and lat pulldown. For each timestep t , the LSTM uses a cell state ct along with a hidden state ht , which are updated using three gates (forget, input, and output). The forget gate decides how much of the previous ct to discard. The input gate decides how much of the current session's information to add. The output gate decides what part of the updated ct is exposed as ht . This gating structure allows the network to remember long range patterns, such as fatigue accumulated over weeks of training or the effects of a previous low volume session, while still updating in response to the most recent session. An LSTM model was chosen over a transformer, as a transformer would require a much larger quantity of data than the datasets being used. The final ht is passed through a small fully connected output layer to produce the predicted weight and reps for the next session.

D. Baselines

The LSTM model will be compared against five separate baselines. The first three will be standard lifting heuristics while the last two will be more simple machine learning models which will test if the complexity of the LSTM model is worth it.

1. **Recent Performance:** Will assume that the next session's top set weight and reps for an exercise will be the same as the last session in which that exercise was performed.
2. **Linear Progression Heuristic:** +2.5kg or +1 rep per week. This is the generic progression metric used by most lifters.
3. **Epley 1RM Formula:** $\text{Weight} * (1 + \frac{\text{reps}}{30})$. This formula is used to estimate a lifter's one rep max from the weight and reps of their top set.
4. **Linear Regression Model:** This model will be trained with the same engineered features as the LSTM model.
5. **Gradient Boosted Trees Model:** This model will be trained with the same engineered features as the LSTM model.

E. Evaluation

To ensure the model is tested fairly and to prevent data leakage, the train, validation, and test splits will be performed in chronological order rather than randomly. This is necessary because the final model must predict a future event in a sequence; training on future sessions would leak information the model would not have access to in a real deployment, inflating apparent performance. Sessions will be split approximately 70% for training, 15% for validation, and 15% for testing, with the

validation set used for hyperparameter tuning and early stopping, and the test set reserved for final, one-time evaluation.

Performance will be measured using root mean squared error (RMSE) and mean absolute error (MAE), computed separately for the bench press and lat pulldown. Given the limited size of the test set, per-session error will also be examined alongside these aggregate metrics, to avoid over-interpreting differences that may be driven by only a small number of sessions.

Feature importance will be examined using each model's native method where available: coefficient magnitudes for linear regression and built in importance scores for the gradient boosted trees model. For the LSTM, which has no equivalent built in measure, feature importance will be estimated via ablation (removing one feature at a time, retraining, and measuring the resulting change in validation error).

IV. EXPECTED RESULTS

The project expects to produce a working next session predictor with quantified error against all baselines, and an analysis of which engineered features contribute most to prediction accuracy, via feature importance. A key possible outcome is that gradient boosted trees perform similarly to the LSTM; this would itself be a meaningful finding, indicating that the temporal sequence structure the LSTM is designed to exploit does not add value beyond well engineered rolling features at this data scale.

V. ETHICAL CONSIDERATIONS

Because prediction errors could encourage over training or an unsafe load increase, this project treats any resulting model as a decision aid rather than a replacement for coaching judgement. The final report will discuss how much confidence a lifter should reasonably place in an automated prediction when the consequence of error includes risk of injury.

STATEMENT

Tools used: Anthropic Claude Sonnet 5 was used to assist in brain storming and structuring for the proposal and report as well as refine wording and grammar. Kaggle and Hevy were used to find the public datasets in Section III-A.

The analysis, decisions and writing submitted in this report are my own.

REFERENCES

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